

Modern Physics Notes

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Abstract

This is a simple abstract for the Modern Physics notes.

1 Introduction

Class test:

10% $\times$ 2 (special relativity + quantum mechanics).

+ 80% exam: 4 short questions (2 + 2) + 2 long questions.

2 Lecture 1: Galilean Relativity

2.1 Reference Frame

It is about Newton's first law: An object at rest stays at rest, and an object in motion stays in motion with the same speed and in the same direction, unless acted upon by a net external force. Clearly, a reference frame that moves relative to an inertial reference frame at a constant speed is also an inertial reference frame.

Definition 2.1

- Inertial reference frame: a reference frame in which Newton's first law holds.
- Non-inertial reference frame: a reference frame in which Newton's first law does not hold.

It is really important because when we use different reference frames to describe the motion of the same object, we may get different results.

2.2 Velocity and Acceleration

Definition 2.2

Velocity is the rate of change of position with respect to time, and acceleration is the rate of change of velocity with respect to time.

$$\begin{aligned} \text{velocity} \quad \vec{v} &= \frac{d\vec{r}}{dt} \\ \text{acceleration} \quad \vec{a} &= \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2} \end{aligned}$$

For different theories, the definitions of velocity and acceleration remain the same, because their definitions are quite simple, but for energy and momentum, the expressions are quite different.

2.3 Time

There is only one thing to mention: in Galilean relativity, time is absolute, so velocity and acceleration depend on change in position vector; the only difference between different reference frames is different representation of the position vector. But in special relativity, time changes.

2.4 Describing Motion in Different Reference Frames

2.4.1 Concepts

Different inertial reference frames give different descriptions of the same event. The position vector and velocity vector depend on the choice of the reference frame, but in Galilean relativity time is absolute, and acceleration is the same in all inertial frames.

In the standard configuration, S' moves along the $+x$ direction with constant velocity $\vec{V}$ relative to S . It does not matter which frame is called S and which is called S' , as long as we are consistent.

Let frame S' move with constant velocity $\vec{V}$ relative to frame S . If the origins coincide at $t = 0$, then the position vectors are related by

$$\vec{r} = \vec{r}' + \vec{V}t, \quad \vec{r}' = \vec{r} - \vec{V}t.$$

Differentiating with respect to time gives the velocity transformation

$$\vec{v}' = \vec{v} - \vec{V}, \quad \vec{v} = \vec{v}' + \vec{V}.$$

Since $\vec{V}$ is constant,

$$\vec{a}' = \vec{a}.$$

In Galilean relativity, time is absolute:

$$t' = t.$$

In the standard configuration, where S' moves along the positive x direction,

$$x' = x - Vt, \quad y' = y, \quad z' = z, \quad t' = t,$$

and

$$v'_x = v_x - V, \quad v'_y = v_y, \quad v'_z = v_z.$$

Remark

In Galilean relativity: time is absolute; position and velocity depend on the reference frame; acceleration is universal in all inertial frames.

The full derivation is given again in Section 2.1.

2.4.2 Original Note

We define the position vector $\vec{r}$ and the velocity vector $\vec{v}$. They depend on the choice of the reference frame.

If we choose different reference frames, the position vector and the velocity vector will be different, related by the following equations:

$$\vec{r} = \vec{r'} + \vec{R}$$

(we assume that the position vector of particle X in frame S is $\vec{r}$ and in frame S' is $\vec{r'}$, and the position vector of the origin of frame S' relative to frame S is $\vec{R}$)

Differentiating, we can get the relation of the velocity vector in different reference frames; overall it is similar to the position vector:

$$\begin{aligned}\vec{v} &= \frac{d\vec{r}}{dt} \\ &= \frac{d\vec{r'}}{dt} + \frac{d\vec{R}}{dt} \\ &= \vec{v'} + \vec{V} \\ \vec{v'} &= \vec{v} - \vec{V} \\ &\text{(inverse relation)}\end{aligned}$$

Remember that in Galilean relativity time is absolute, $t' = t$, which allows us to build this relationship. The relation of the position vector is similar to the relation of velocity, as said, “velocity and acceleration depend on change in position vector”.

3 Lecture 2: Galilean Relativity and the Michelson–Morley Experiment

Remark

Relativity is the relationship of one reference frame to another.

3.1 Review of Galilean Transformation

3.1.1 Relations

Let frame S' move with constant velocity $\vec{V}$ relative to frame S . For a particle, denote its position by $\vec{r}$ in S and $\vec{r'}$ in S' . If the origins coincide at $t = 0$, then

$$\vec{R} = \vec{V}t,$$

and

$$\vec{r} = \vec{r'} + \vec{R} = \vec{r'} + \vec{V}t.$$

Therefore

$$\boxed{\vec{r'} = \vec{r} - \vec{V}t.}$$

Differentiating with respect to time,

$$\vec{v}' = \frac{d\vec{r}'}{dt} = \vec{v} - \vec{V},$$

or

$$\boxed{\vec{v}' = \vec{v} - \vec{V}.}$$

The inverse transformation is

$$\boxed{\vec{v} = \vec{v}' + \vec{V}.}$$

Since $\vec{V}$ is constant,

$$\frac{d\vec{V}}{dt} = 0,$$

so

$$\boxed{\vec{a}' = \vec{a}.}$$

If S' is non-inertial, i.e. $\vec{V} = \vec{V}(t)$, then

$$\vec{a}' = \vec{a} - \vec{A}, \quad \vec{A} = \frac{d\vec{V}}{dt}.$$

And finally,

Remark

In Galilean relativity, $t' = t$; time is absolute.

3.1.2 Original Note and Example

A frame S' moves with constant velocity $\vec{V}$ relative to another frame S . According to Galilean relativity,

$$\begin{aligned} \vec{R} &= \vec{V}t \\ \vec{r}' &= \vec{r} - \vec{R} \\ \dot{\vec{R}} &= \vec{V} = \text{const} \\ \vec{r}' &= \vec{r} - \vec{R} = \vec{r} - \vec{V}t \end{aligned}$$

The position transformation does not change time.

$$\begin{aligned} t' &= t \\ \vec{v}' &= \vec{v} - \vec{V} \\ \vec{a}' &= \vec{a} - \vec{A} \end{aligned}$$

Where $\vec{a}'$ is the acceleration in frame S' , $\vec{a}$ is the acceleration in frame S , and $\vec{A} = d\vec{V}/dt$ is the acceleration of frame S' relative to frame S .

In inertial frames, $\vec{A} = 0$, so $\vec{a}' = \vec{a}$.

Remark

In Galilean relativity, acceleration is universal.

For example, on a ship moving with constant velocity, a ball dropped from the crow's nest falls straight down relative to the ship, even though the ship is moving relative to the ground.

3.2 Introduction to the Michelson–Morley Experiment

- At the beginning of the 1800s, Young’s interference experiment demonstrated the wave nature of light.
- A wave requires a medium for propagation, leading to the hypothesis of the luminiferous aether.

Galilean relativity works well for mechanics, but it is not compatible with Maxwell’s electromagnetism. Maxwell’s equations give a fixed speed of light c , and experiments always measured the same speed of light, independent of the motion of the Earth.

At that time, physicists assumed light needed a medium called the luminiferous aether. The Michelson–Morley experiment was designed to detect the Earth’s motion relative to this aether.

3.2.1 Hand-Clapping Experiment

In this experiment, two people stand a certain distance apart and clap their hands when they hear another’s clap.

- case 1: platform is stationary relative to the ground.
- case 2: platform is moving relative to the ground.
- case 3: the line joining the two people is perpendicular to the direction of motion.

For case 1

$$t = \frac{2L}{v_{\text{sound}}}$$

For case 2

$$t = \frac{2Lv_{\text{sound}}}{v_{\text{sound}}^2 - v_{\text{platform}}^2}$$

For case 3

$$t = \frac{2L}{\sqrt{v_{\text{sound}}^2 - v_{\text{platform}}^2}}$$

Clearly,

$$t_2 \geq t_3.$$

As a scientific check, when $v_{\text{platform}} = 0$, we recover

$$t = \frac{2L}{v_{\text{sound}}}.$$

3.2.2 Michelson–Morley Experiment

The main idea is not to measure T_1 and T_2 individually; instead, we use a beam splitter.

We look at the interference pattern. If the two travel times are different, the waves are out of phase, and the interference pattern will shift.

If no shift takes place, it means the two travel times are equal. No aether was detected.

$$T_1 = t_f + t_b = \frac{2Lc}{c^2 - v^2} \quad (\text{parallel arm})$$

$$T_2 = \frac{2L}{\sqrt{c^2 - v^2}} \quad (\text{perpendicular arm})$$

$$T_2 = T_1 \sqrt{1 - \frac{v^2}{c^2}}$$

For the parallel arm, the light travels with speed $c + v$ in one direction and $c - v$ in the other, so

$$T_1 = \frac{L}{c + v} + \frac{L}{c - v} = \frac{2Lc}{c^2 - v^2}.$$

For the perpendicular arm, the light travels along the hypotenuse of a right triangle, so

$$\left(\frac{cT_2}{2}\right)^2 = L^2 + \left(\frac{vT_2}{2}\right)^2,$$

which gives

$$T_2 = \frac{2L}{\sqrt{c^2 - v^2}}.$$

Lorentz and FitzGerald proposed that objects contract along the direction of motion, by the factor $\sqrt{1 - v^2/c^2}$ (or $1/\gamma$), in order to explain the null result.

$$\sqrt{1 - \frac{v^2}{c^2}}$$

This was mathematically close to the correct transformation, but for the wrong physical reason. Einstein later explained the null result by postulating that the speed of light is the same in all inertial frames.

4 Lecture 3

4.1 Natural Units

Definition 4.1

An electron accelerated through a potential difference of 1 V gains kinetic energy 1 eV.

4.2 Van de Graaff Accelerator

Plotting the speed and kinetic energy of electrons using experimental data shows that the speed of electrons approaches a significant fraction of the speed of light.

This shows that the classical formula $T = \frac{1}{2}mv^2$ is wrong at high speeds.

Remark

The speed of light is always c ; this is an experimental fact.

4.3 The Postulates of Special Relativity

4.3.1 Postulate 1

Postulate 1: The laws of physics are the same in all inertial frames; all inertial (non-accelerating) frames are “equivalent”. It is the same as the familiar principle of Galilean relativity.

4.3.2 Postulate 2

Postulate 2: The speed of light in a vacuum is the same value c for any inertial frame. The Michelson–Morley experiment supported this postulate; it was unable to detect any difference in the speed of light due to the motion of the Earth through the supposed aether. There is a limiting speed for any object in any frame.

4.3.3 Units of Measurement

Definition 4.2

- Speed of light: $c = 299792458$ m/s.
- Second: defined by the hyperfine transition frequency of Cs-133.
- Meter: defined by the distance light travels in vacuum in $1/299792458$ seconds.

4.4 Transverse Rods

Consider rod A moving perpendicular to rod B of the same proper length. Suppose the transverse length changed by a factor $f(v)$, so that A would measure B’s transverse length as $f(v)L$, and B would measure A’s transverse length as $f(v)L$. By symmetry $f(v) = f(-v)$. But the transverse direction has no preferred orientation, so no such factor can appear. Therefore $f(v) = 1$, and the length of a rod moving perpendicular to the direction of motion remains unchanged.

Remark

Perpendicular length remains unchanged.

4.5 The Role of Observers

The act of measuring should not interfere with what we are measuring.

4.6 The Frame of Reference in Special Relativity

- relativity of simultaneity
- time dilation
- length contraction

We need to define time at different points in a consistent manner for all observers.

5 Lecture 4: Introduction to Special Relativity

5.1 Time in Special Relativity

In special relativity, time is not absolute and can vary depending on the relative motion of observers. Every coordinate has its own time. We can use four-vector notation (x, y, z, ct) to represent spacetime events.

5.2 Synchronising Clocks

A light source sends photons from the middle of the rod; in the rest frame of the rod, the photons reach both ends simultaneously. This allows observers to synchronise clocks at both ends of the rod accurately.

This defines simultaneity in that frame.

5.3 Events

Definition 5.1

An event is where and when something happens.

A reference frame provides the coordinate system and clocks used to assign position and time to events.

Relativity deals with events in (x, y, z, t) .

5.4 Proper Time

Definition 5.2

The time interval between two events within the same spatial coordinates is called the proper time.

All other time intervals are improper time intervals.

If, in a given reference frame, the two events occur at different spatial positions, then the time interval measured in that frame is improper.

5.5 Time Dilation

Definition 5.3

Time dilation

$$\tau = \gamma \tau_0 \quad \text{where } \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$$

- proper time τ_0 : the time interval between two events occurring at the same spatial location.
- dilated time τ : the time interval measured in a frame where the clock is moving.
- Lorentz factor $\gamma = 1/\sqrt{1 - v^2/c^2}$.

- scenario 1: Observer and clock are stationary relative to each other.
- scenario 2: Observer and clock are moving relative to each other.

$$\begin{aligned}\tau_0 &: \text{proper time (scenario 1)} \\ \tau &= \gamma\tau_0 \quad (\text{scenario 2}) \\ \gamma &= \frac{1}{\sqrt{1 - v^2/c^2}} \quad (\text{Lorentz factor})\end{aligned}$$

5.5.1 Light Bouncing Between Mirrors

A platform carrying two mirrors, one at the top and one at the bottom, moves relative to the ground frame S. We have three events:

- Event 1: Light leaves the source.
- Event 2: Light reaches the top mirror.
- Event 3: Light reaches the bottom mirror.

For the platform frame S', the coordinates of the events are as follows:

<i>Event</i>	<i>x</i>	<i>y</i>	<i>z</i>	<i>t</i>
1	0	0	0	0
2	0	l_0	0	l_0/c
3	0	0	0	$2l_0/c$

The time interval between events 1 and 3 in S' is $2l_0/c$. It is proper time because the position of the clock (light source) does not change in S'.

$$\Delta t'_{13} = \tau_0 = \frac{2l_0}{c}$$

When observed from the ground frame S, the light travels a longer diagonal path due to the motion of the platform.

The length of the diagonal path traveled by the light is l_{eff} . The three events are listed as follows in the ground frame S:

<i>Event</i>	<i>x</i>	<i>y</i>	<i>z</i>	<i>t</i>
1	0	0	0	0
2	vl_{eff}/c	l_0	0	l_{eff}/c
3	$2vl_{\text{eff}}/c$	0	0	$2l_{\text{eff}}/c$

The y -coordinate is unchanged because the motion is along x .

The time interval between events 1 and 3 in S is $2l_{\text{eff}}/c$; it is improper time because the clock (light source) is moving in S.

$$\Delta t_{13} = \frac{2l_{\text{eff}}}{c} = \tau$$

Because the light travels in the long diagonal path in the ground frame S, we can use the Pythagorean theorem to relate l_{eff} and l_0 , and the distance traveled due to the platform's motion:

$$l_{\text{eff}}^2 = l_0^2 + \left(v \frac{l_{\text{eff}}}{c}\right)^2$$

Solving for l_{eff} :

$$l_{\text{eff}} = \frac{l_0}{\sqrt{1 - v^2/c^2}} = \gamma l_0$$

And also we can relate the time intervals in the two frames, using the Pythagorean theorem:

$$l_0^2 + \left(\frac{v\tau}{2}\right)^2 = \left(\frac{c\tau}{2}\right)^2$$

Solving for τ :

$$\tau = \frac{2l_0}{\sqrt{c^2 - v^2}} = \frac{2l_0}{c\sqrt{1 - v^2/c^2}}$$

$$\tau = \gamma \frac{2l_0}{c}$$

$$\tau_0 = \frac{2l_0}{c}$$

$$\tau = \gamma \tau_0$$

5.6 The Lorentz Factor γ

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$$

- γ is the Lorentz factor. Time intervals are multiplied by γ , lengths along the direction of motion are divided by γ , and relativistic momentum and energy involve γ .
- By definition, $\gamma \geq 1$.
- γ becomes very large as v approaches c , indicating significant relativistic effects.
- When $\gamma = 1$, the relative velocity $v = 0$, indicating no relativistic effects.

5.6.1 On Time Dilation

- Proper time is the time interval between two events occurring at the same spatial location.
- Time dilation occurs when observing a moving clock, which appears to run slower than a stationary clock.
- Proper time: τ_0 . Dilated time: $\tau = \gamma \tau_0$.

If A and B move relative to each other, each observes the other's clock running slower. This is not a contradiction; it depends on which frame is used.

5.6.2 Experimental Proof

We use the muon lifetime as an experimental proof of time dilation.

Muons are produced in the upper atmosphere and travel towards the Earth's surface. Due to their short lifetime, without time dilation, very few muons would reach the Earth's surface.

However, because of time dilation, moving muons experience a longer lifetime from the perspective of an observer on Earth, allowing many of them to reach the surface.

Use t_0 as the half-life of the muon at rest, for a moving muon, the effective half-life is $t_{1/2} = \gamma t_0$.

And for a radioactive sample,

$$\begin{aligned}-\frac{dN}{dt} &\propto N \\ N(t) &= N_0 e^{-t/\tau_{\text{mean}}} \\ t_{1/2} &= \tau_{\text{mean}} \ln 2\end{aligned}$$

Setting $N(t_{1/2}) = N_0/2$ gives $t_{1/2} = \tau_{\text{mean}} \ln 2$.

For a moving muon, the classically expected surviving fraction is $P = 0.05$, but experimentally $P = 0.72 \pm 0.03$.

This discrepancy between the classical expectation and the experimental observation is explained by time dilation, which increases the effective lifetime of moving muons as observed from the Earth's frame.

6 Lecture 5: Space and time in Special Relativity

6.1 Lorentz Contraction

6.1.1 Definition

Definition 6.1

Lorentz contraction

$$L = L_0 \frac{1}{\gamma} \quad \text{where } \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$$

1. proper length L_0 : event time coordinates are the same when measuring the length of an object, length in the system at rest
2. improper length L : length when viewed moving at speed v

Example

Length Contraction of a Muon

In the muon frame, the distance between the muon and earth is Lorentz contracted compared to the distance measured in the Earth's frame.

6.1.2 Thought experiment: the moving train

The train is moving at a velocity v relative to an observer on the platform.

Proper length

Observer in the middle of the platform sends two flashes of light (same technique used to synchronise clocks)

- $(x_1 = 0, t_1 = T)$ the light reaches the first end of the platform
- $(x_2 = 2cT, t_2 = T)$ the light reaches the far end of the platform

Length of the platform is $l_0 = 2cT$, because the events occur at the same time in the platform frame, $t_1 = t_2$ so l_0 is the proper length.

In the Platform Frame

- $(0, t_1)$ when the train passes at one end of the platform
- (l, t_2) when the train passes at the far end of the platform

the time interval is

$$\Delta t = t_2 - t_1 = \tau$$

with different space coordinates

$$x'_1 \neq x'_2 \Rightarrow \tau \text{ not a proper time: the events are seen at the same coordinate}$$

And the observer on the platform confirms that

$$l_0 = v\tau$$

In the train driver frame of reference

The driver is not moving while the platform is passing at a speed v .

- Event 1, corresponding to the near end of the platform: $(x'_1 = 0, t'_1)$
- Event 2, corresponding to the far end of the platform: $(x'_2 = l, t'_2)$

The time interval is

$$\Delta t = t'_2 - t'_1 = \tau_0$$

$$x'_1 \neq x'_2 \Rightarrow \tau_0 \text{ is a proper time: the events are seen at the same coordinate}$$

The deduced length of the platform is:

$$\Delta x' = l = v\tau_0$$

$$t'_1 \neq t'_2 \Rightarrow l \text{ is an improper length: the events are not seen at the same time}$$

Length contraction

the two lengths determined above are

$$\begin{aligned}l_0 &= v\tau \\ l &= v\tau_0\end{aligned}$$

The ratio between the lengths observed by the two observers is

$$\begin{aligned}\frac{l}{l_0} &= \frac{v\tau_0}{v\tau} \\ \tau &= \gamma\tau_0 \\ l &= \frac{l_0}{\gamma}\end{aligned}$$

And length contraction takes place. This is equivalent to the time dilation effect.

6.1.3 The twin paradox

Twin A on earth remains stationary, while Twin B travels on a spaceship at a high velocity.

1. From A's perspective, Twin B's clock runs slower due to time dilation. As a result, when Twin B returns to Earth, they will have aged less than Twin A, despite both experiencing the passage of time according to their own frames of reference.
2. In B's frame, it is Twin A who appears to be moving, and thus Twin A's clock runs slower from Twin B's perspective. However, the asymmetry arises because Twin B undergoes acceleration and deceleration during the journey, changing inertial frames, while Twin A remains in a single inertial frame. This explains why Twin B ages less upon returning to Earth.

The error lies in the fact that B doesn't remain in the same inertial frame throughout the journey, while A does. B accelerated while changing direction, which breaks the symmetry and results in B aging less than A upon reunion.

6.2 Moving Synchronised Clocks

Consider two clocks that are synchronised in their own rest frame and are moving with a constant velocity relative to an observer. Let frame S move with respect to the page, and let frame S' be the page.

In the rest frame of the clocks, the two clocks are synchronised, and their separation is Δx . In the page frame S' , the two clocks are moving with speed v , and the distance between them is contracted:

$$AM = BM = \frac{\Delta x}{2\gamma}.$$

The light signals emitted from the midpoint reach the two clocks at

$$\begin{aligned}ct'_A &= \frac{\Delta x}{2\gamma} - vt'_A \implies t'_A = \frac{\Delta x}{2\gamma(c + v)}, \\ ct'_B &= \frac{\Delta x}{2\gamma} + vt'_B \implies t'_B = \frac{\Delta x}{2\gamma(c - v)}.\end{aligned}$$

Therefore,

$$t'_A \neq t'_B \implies \Delta t' = t'_B - t'_A = \frac{\gamma v \Delta x}{c^2}.$$

Since $t'_A < t'_B$, the leading clock lags behind the trailing clock.

Definition 6.2

Leading clock lags (clock lag)

If two clocks are synchronised in their rest frame and move with velocity v relative to an observer, the leading clock lags behind the trailing clock by

$$\Delta t = \frac{\gamma v L}{c^2},$$

where $L = \Delta x$ is the proper distance between the two clocks.

Remark

The clocks are not synchronised when viewed from a frame moving with a different speed. This is a direct consequence of the relativity of simultaneity.

6.2.1 The rod-Shed Paradox

The rod-shed paradox involves a rod moving at a high velocity towards a shed. In the shed's frame, the rod undergoes length contraction and can fit entirely within the shed. However, in the rod's frame, the shed is moving and undergoes length contraction, making it appear too short to contain the rod. The length of shed is L_0 , the length of the rod is l_0 .

$$\begin{aligned} l_0 &= 2L_0 \\ v &= \frac{\sqrt{3}}{2}c \\ \gamma &= \frac{1}{\sqrt{1 - v^2/c^2}} = 2 \end{aligned}$$

Shed frame

$$\begin{aligned} L &= L_0 \\ l &= \frac{l_0}{\gamma} = \frac{2L_0}{2} = L_0 \end{aligned}$$

The rod can be fully contained within the shed in the shed's frame due to length contraction.

Rod frame

$$\begin{aligned} l &= l_0 \\ L &= \frac{L_0}{\gamma} = \frac{L_0}{2} \end{aligned}$$

The shed is length contracted and appears too short to contain the rod.

In the rod's frame, clock D leads clock C.

1. C reaches A $\rightarrow$ door C closes
2. D reaches B $\rightarrow$ door D closes

3. Door D closed after door C: “The leading clock lags”

The synchronised clocks in the rod’s frame are not simultaneous with those in the shed’s frame due to the relativity of simultaneity.

6.2.2 Transverse and Parallel Motion

From the perspective of leading clock lag.

1. In the transverse motion, the proper length can consistently be measured: both ends can always be measured at the same time, because the motion is perpendicular to the direction of measurement.
2. In the parallel motion, however, the proper length cannot be measured simultaneously in the moving frame because the motion is along the direction of measurement, since the leading clock lags, leading to length contraction.

6.3 Summary

- time dilation: “a moving clock runs slow”

$$\tau = \gamma\tau_0$$

- space contraction: “a moving stick is shorter”

$$l = \frac{l_0}{\gamma}$$

- synchronisation: “a leading clock lags”

$$\Delta t = \frac{\gamma v L}{c^2}$$